

ACR42701

## Hydrogen chloride, 1.25M solution in ethanol

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
Product Description: 氯化氢, 1.25M乙醇溶液  
Hydrogen chloride, 1.25M solution in ethanol

Cat No. : 427010000; 427011000; 427018000

Supplier **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
No information available

**Odor**  
No information available

**Emergency Overview**  
Highly flammable liquid and vapor. Causes severe skin burns and eye damage.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Flammable liquids.                | Category 2   |
| Skin Corrosion/Irritation         | Category 1 A |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



Signal Word

Danger

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

**Hazard Statements**

H225 - Highly flammable liquid and vapor

H314 - Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Highly flammable. Vapors may cause flash fire or explosion.

**Health Hazards**

Corrosive. Causes skin and eye burns.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . The product is water soluble, and may spread in water systems.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Ethyl alcohol     | 64-17-5   | 93 - 96  |
| Hydrogen chloride | 7647-01-0 | 4 - 7    |

**SECTION 4. FIRST AID MEASURES****General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

### Most important symptoms and effects

Difficulty in breathing. Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Notes to Physician

Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe mist/vapors/spray. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Storage

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from heat, sparks and flame. To maintain product quality: Refrigerator/flammables.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component         | China                          | Taiwan                                       | Thailand       | Hong Kong                                        |
|-------------------|--------------------------------|----------------------------------------------|----------------|--------------------------------------------------|
| Ethyl alcohol     | -                              | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | TWA: 1000 ppm  | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>     |
| Hydrogen chloride | Ceiling: 7.5 mg/m <sup>3</sup> | -                                            | Ceiling: 5 ppm | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup> |

| Component         | ACGIH TLV      | OSHA PEL                                                                                                             | NIOSH                                                          | The United Kingdom                                                                                         | European Union                                                                                                 |
|-------------------|----------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol     | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup>     | IDLH: 3300 ppm<br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm STEL; 5760 mg/m <sup>3</sup> STEL |                                                                                                                |
| Hydrogen chloride | Ceiling: 2 ppm | Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 ppm<br>(Vacated) Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | STEL: 5 ppm 15 min<br>STEL: 8 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 2 mg/m <sup>3</sup> 8 hr | TWA: 5 ppm (8h)<br>TWA: 8 mg/m <sup>3</sup> (8h)<br>STEL: 10 ppm (15min)<br>STEL: 15 mg/m <sup>3</sup> (15min) |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Butyl rubber   | See manufacturers | -               |             | (minimum requirement) |

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

|                 |        |
|-----------------|--------|
| recommendations | EN 374 |
|-----------------|--------|

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                   |                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>   | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>     | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>  | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Acid gases filter Type E Yellow or Organic gases and vapours filter Type A Brown conforming to EN14387                               |
| <b>Small scale/Laboratory use</b> | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Appearance</b>                              |                          |                                             |
| <b>Physical State</b>                          | Liquid                   |                                             |
| <b>Odor</b>                                    | No information available |                                             |
| <b>Odor Threshold</b>                          | No data available        |                                             |
| <b>pH</b>                                      | No information available |                                             |
| <b>Melting Point/Range</b>                     | No data available        |                                             |
| <b>Softening Point</b>                         | No data available        |                                             |
| <b>Boiling Point/Range</b>                     | No information available |                                             |
| <b>Flash Point</b>                             | 12 °C / 53.6 °F          | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available        |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available        |                                             |
| <b>Vapor Pressure</b>                          | No data available        |                                             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.825                    |                                             |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | No information available |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Ethyl alcohol                                  | -0.32                    |                                             |
| <b>Autoignition Temperature</b>                | No data available        |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | No data available        |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

**Oxidizing Properties** No information available

### SECTION 10. STABILITY AND REACTIVITY

**Stability** Stable under normal conditions. Hygroscopic.

**Hazardous Reactions** None under normal processing.  
**Hazardous Polymerization** Hazardous polymerization does not occur.

**Conditions to Avoid** Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

**Materials to avoid** Strong oxidizing agents.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

(a) acute toxicity;  
 Toxicology data for the components

| Component         | LD50 Oral                                                    | LD50 Dermal             | LC50 Inhalation                                                                                                                   |
|-------------------|--------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol     | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) |                         | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat)                                                                 |
| Hydrogen chloride | 900 mg/kg ( Rabbit )                                         | > 5010 mg/kg ( Rabbit ) | LC50 = 4701 ppm (rat) 30 min (gas), LC50 = 588 ppm (4h) by extrapolation<br>LC50 = 8.3 mg/L (rat ) 30 min (aerosols) (MMAD < 5µm) |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
**Respiratory** No data available  
**Skin** No data available

| Component                            | Test method                                                | Test species | Study result    |
|--------------------------------------|------------------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 93 - 96 ) | Mouse Ear Swelling Test (MEST)                             | mouse        | non-sensitising |
|                                      | -----<br>OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                            | Test method                                            | Test species         | Study result |
|--------------------------------------|--------------------------------------------------------|----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 93 - 96 ) | AMES test<br>OECD Test Guideline 471                   | in vitro<br>Bacteria | negative     |
|                                      | -----<br>Gene cell mutation<br>OECD Test Guideline 476 | in vitro             | negative     |

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

|  |  |           |  |
|--|--|-----------|--|
|  |  | Mammalian |  |
|--|--|-----------|--|

**(f) carcinogenicity;** No data available

Ethanol has been shown to be carcinogenic in long-term studies only when consumed and abused as an alcoholic beverage. The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available

| Component                            | Test method             | Test species / Duration   | Study result          |
|--------------------------------------|-------------------------|---------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 ( 93 - 96 ) | OECD Test Guideline 416 | Oral / mouse 2 Generation | NOAEL = 13.8 g/kg/day |
|                                      | OECD Test Guideline 414 | Inhalation / Rat          | NOAEC = 16000 ppm     |

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. Do not flush into surface water or sanitary sewer system.

| Component     | Freshwater Fish                                                  | Water Flea                                    | Freshwater Algae                              | Microtox                                                                                                                |
|---------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |

### Persistence and Degradability

**Persistence** Persistence is unlikely.

| Component                            | Degradability   |
|--------------------------------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 93 - 96 ) | OECD 301E = 94% |

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Ethyl alcohol | -0.32   | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

**SAFETY DATA SHEET**

Hydrogen chloride, 1.25M solution in ethanol

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN2924  
 Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
 Technical Shipping Name Contains Ethanol, Hydrogen chloride  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

**IMDG/IMO**

UN-No UN2924  
 Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
 Technical Shipping Name Contains Ethanol, Hydrogen chloride  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

**IATA**

UN-No UN2924  
 Proper Shipping Name Flammable liquid, corrosive, n.o.s.  
 Technical Shipping Name Contains Ethanol, Hydrogen chloride  
 Hazard Class 3  
 Subsidiary Hazard Class 8  
 Packing Group II

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Ethyl alcohol     | X                                                   | X                                       | X    | X     | 200-578-6 | X    | X   | X     | X    | X    | X    | KE-13217 |
| Hydrogen chloride | X                                                   | X                                       | X    | X     | 231-595-7 | X    | X   | X     | X    | X    | X    | KE-20189 |

# SAFETY DATA SHEET

## Hydrogen chloride, 1.25M solution in ethanol

| Component         | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Hydrogen chloride | 25 tonne                                                                                  | 250 tonne                                                                                |

### National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 02-Jun-2009  
**Revision Date** 11-Apr-2024  
**Revision Summary** Not applicable.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards** On basis of test data  
**Health Hazards** Calculation method  
**Environmental hazards** Calculation method

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

# SAFETY DATA SHEET

Hydrogen chloride, 1.25M solution in ethanol

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**